

THD+Noise Estimation in Class-D Amplifiers

Burak Kelleci

Edgar Sánchez-Sinencio

and Aydın İlker Karşılayan

Department of Electrical and Computer Engineering

Texas A&M University

College Station, Texas 77843-3128

Email: burak@ece.tamu.edu

Abstract—A mathematical analysis of the effect of jitter and nonlinear inductance on THD+Noise of class-D amplifiers is presented. In the analysis, it is assumed that jitter is not correlated between edges and has Gaussian distribution. It is shown that the nonlinear inductor in the output filter causes THD+Noise to be frequency dependent. The relationship between jitter, nonlinear inductor and circuit parameters is formulated. The accuracy of the proposed mathematical approximation is verified by transistor level simulations.

I. INTRODUCTION

The drive of today's consumer market towards portability and style in audio products has given an old idea, the class-D amplifier, a new stage where it can compete with Class A and AB amplifiers. The efficiency of a class-D amplifier is theoretically 100%, and its THD is comparable to class AB and A amplifiers. High efficiency is obtained by using a switched output stage, therefore audio information has to be inserted by using special modulation techniques. A widely-used technique is pulse-width modulation (PWM). There are mainly two techniques to generate PWM signal. One is based on sigma-delta modulation [1] and the other is comparing the input signal with a triangular or saw-tooth wave [2]. The triangular technique, as shown in Fig. 1, is widely used in the industry. The comparator generates PWM signal by comparing the input signal with an internally generated triangular wave. The output transistors operate as switches and are used to drive low impedance load. Since PWM signal includes spectral components at the signal and triangular wave frequencies, the output signal must be filtered using a passive and lossless filter, providing maximum power transfer with high efficiency.

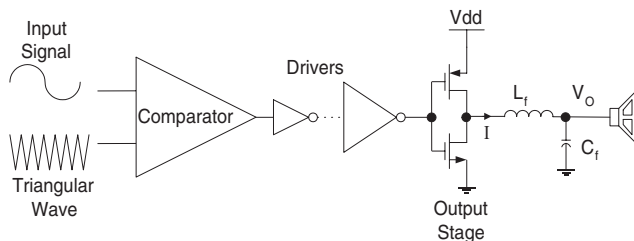


Fig. 1. Basic Class-D Amplifier Topology

The jitter is the uncertainty at the edges of pulse width modulated (PWM) signal. Main sources of jitter are phase

noise of the wave generators, thermal and flicker noise of the input signal, and threshold variation in the comparator, drivers and output switches. Since the information is modulated using the position of edges, time jitter causes information loss. The theoretical derivations of PWM spectra assume square edges, and this will be unrealistic for any practical circuits. The finite rise and fall times is considered as simply low pass filtering of the ideal PWM signal [3]. However, this assumption is not valid when the pulse duration is shorter than sum of rise and fall times. In that case, the signal will be clipped, and distortion components start to arise.

For performance measurement of class-D amplifiers, total harmonic distortion (THD) and signal-to-noise ratio (SNR) are usually combined as THD+Noise. THD is defined as the ratio of the sum of all harmonics' power within the audio band to the fundamental signal power, whereas SNR is the ratio of signal power to the noise power in the audio band. The effect of the output filter on THD is examined in [4], where Gaussian type filters are found to be suitable for achieving the lowest THD. The effect of nonlinear triangular wave and on-resistance of the output stage is investigated in [5]. It is shown that the on-resistance at the output stage has little effect on THD. In addition, a mathematical analysis method to model the non-linearity of the triangular wave is provided. The randomization of the edges can improve the THD performance of PWM modulator. This technique is examined in the literature for uniform distribution [6], [7].

In this paper two important parameters in the class-D amplifier that contribute to THD+Noise are investigated: cycle-to-cycle jitter at the edges of PWM signal and nonlinearity of the inductor in the output filter. The proposed method provides mathematical relationships among these effects and circuit parameters. This formulation gives good insight to the designer at the early stage of development.

II. SNR AND THD ESTIMATION

The jitter is modeled by

$$y(t) = x(t + j(t)) \quad (1)$$

where $x(t)$ is a jitter-free signal and $j(t)$ is a zero-mean uncorrelated Gaussian process [8]. Since Gaussian distribution is not bounded, it is a finite possibility that jitter term exceed the pulse period. However, this probability is very small for

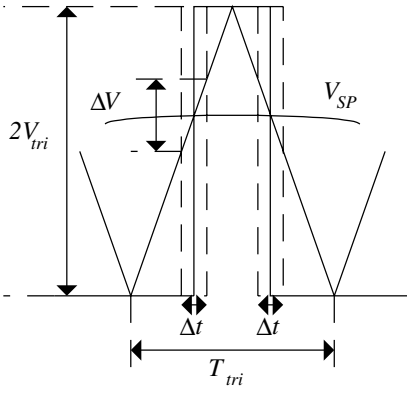


Fig. 2. Equivalent Noise Voltage Model of Jitter

practical purposes. Therefore, it is assumed that jitter is much smaller than the pulse period, and does not cause total loss of the pulse. Using the equivalent noise voltage model [9], it is represented as an input-referred noise voltage, as illustrated in Fig. 2:

$$\Delta V = \frac{2V_{tri}}{T_{tri}/2} \Delta t \quad (2)$$

where Δt is rms of jitter, ΔV is input referred noise voltage, V_{tri} and T_{tri} are the amplitude and the period of the triangular wave, respectively. Since jitter is a Gaussian process and there is a linear relationship between Δt and ΔV , the distribution of ΔV is also Gaussian. Therefore, ΔV shows white noise spectrum between $-1/(2T_{tri})$ and $1/(2T_{tri})$, and its average power equals to ΔV^2 . For a sinusoidal input, input-referred SNR is calculated using Eq. (2) as

$$\begin{aligned} \text{SNR}_{\text{input}} &= \frac{V_{SP}/\sqrt{2}}{\Delta V \sqrt{f_{bw} T_{tri}}} \\ &= \frac{V_{SP}}{4\sqrt{2}\Delta t V_{tri}} \sqrt{\frac{T_{tri}}{f_{bw}}} \end{aligned} \quad (3)$$

where V_{SP} is the input signal peak amplitude and f_{bw} is the audio bandwidth. Output SNR is proportional to the input SNR [10]-[11], and is calculated as

$$\begin{aligned} \text{SNR}_{\text{output}} &= K \cdot \text{SNR}_{\text{input}} \\ &= K \cdot \frac{V_{SP}}{4\sqrt{2}\Delta t V_{tri}} \sqrt{\frac{T_{tri}}{f_{bw}}} \end{aligned} \quad (4)$$

where K is the proportionality factor and usually equals to one. Since in class-D amplifiers THD+Noise is measured, Eq. 4 is rewritten

$$\begin{aligned} (\text{THD+Noise})_{\text{output}} &= \frac{100}{\text{SNR}_{\text{output}}} \\ &= 100 \frac{4\sqrt{2}V_{tri}\Delta t}{V_{SP}} \sqrt{f_{tri} \cdot f_{bw}} \end{aligned} \quad (5)$$

where THD+Noise is in percentage and f_{tri} is triangular wave frequency.

THD performance of PWM signal is inherently independent of the input signal frequency. However, class-D amplifiers usually drive inductive load, such as inductance of a low pass filter. The low saturation current of the load inductor causes frequency dependent nonlinearity. Saturation current of inductor is defined as the I_{DC} current level where the effective inductance value is decreased to 90% percent of its value at zero DC current. An approximate expression of nonlinear inductance is obtained by series representation. Since compression is caused by the odd order terms, even order terms are taken zero. To simplify further, higher than third order terms is also neglected. As a result, the following model is derived,

$$\begin{aligned} L &= L_0 (1 - a_3 I^2) \\ &= L_0 \left(1 - \frac{0.1}{I_{sat}^2} I^2 \right) \end{aligned} \quad (6)$$

where I_{sat} is the saturation current and L_0 is the inductance at zero DC current. At lower frequencies than cutoff frequency of the filter, input signal is approximately equal to the output signal. Assuming weak nonlinearity, the dominant harmonic will be the third harmonic. Therefore, THD is given as,

$$\begin{aligned} \text{THD (in \%)} &= 100 \frac{V_{\text{third harmonic}}}{V_{\text{fundamental}}} \\ &= 100 \frac{V_0^3 \frac{L_0 a_3}{R^3} \frac{3 \cdot 2\pi f_{in}}{4}}{V_0} \\ &= 100 \frac{3 \cdot L_0 V_0^2 \cdot 0.1 \cdot 2\pi f_{in}}{4 \cdot R^3 I_{sat}^2} \end{aligned} \quad (7)$$

where V_0 is the amplitude of the output signal, R is the resistance of the speaker, f_{in} is the input signal frequency. Eq. (7) is valid when the input signal frequency is lower than the cutoff frequency. The derivation of Eq. 7 is explained in appendix.

III. SIMULATION RESULTS

The class-D amplifier in Fig. 1 is designed in $0.35\mu\text{m}$ CMOS technology. The simulations are performed using BSIM3v3 models in Cadence. The comparator is a self-biased differential amplifier [12], which is shown in Fig. 3. The output driver is an inverter chain. The triangular wave generator is implemented as a macromodel to model the jitter. Triangular wave frequency is selected as 1 MHz, so in-band intermodulation products are negligible. The sum of fall and rise times of the output stage is 100ns. Triangular wave amplitude is chosen twice of the input signal amplitude to avoid distortion due to clipping.

Figure 4 shows THD+Noise vs. jitter, where dashed line is the theoretical result and solid line is simulation. Figure 5 shows THD+Noise vs. input signal amplitude, where triangular wave frequency and jitter are selected as 1 MHz and 1 ns, respectively. Input signal amplitude is swept from 1/1000 to 1 of the triangular wave amplitude. THD+Noise in Fig. 5 starts to increase above $0.3V_{tri}$ and it increases dramatically above $0.9V_{tri}$ due to the clipping. Triangular wave amplitude sweep

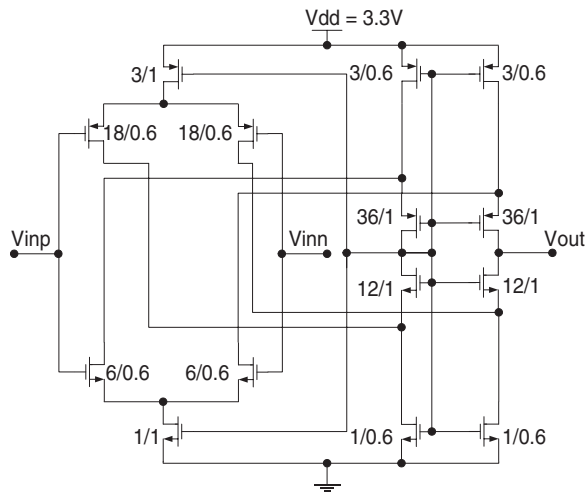


Fig. 3. Comparator Schematic [12]

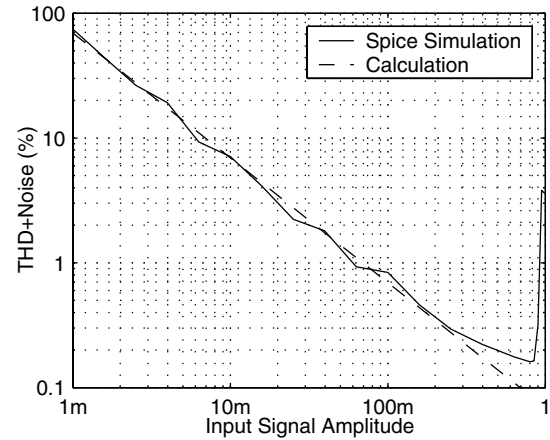


Fig. 5. THD+Noise vs. Signal Amplitude [$V_{tri} = 1V$, $\Delta t = 1ns$, $T_{tri} = 1\mu s$, $f_{in} = 1KHz$, $f_{bw} = 15 KHz$]

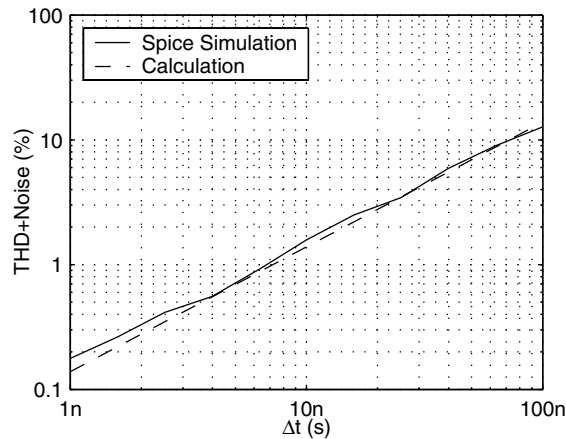


Fig. 4. THD+Noise vs. Jitter [$V_{tri} = 1V$, $V_{SP} = 0.5V$, $T_{tri} = 1\mu s$, $f_{in} = 1KHz$, $f_{bw} = 15 KHz$]

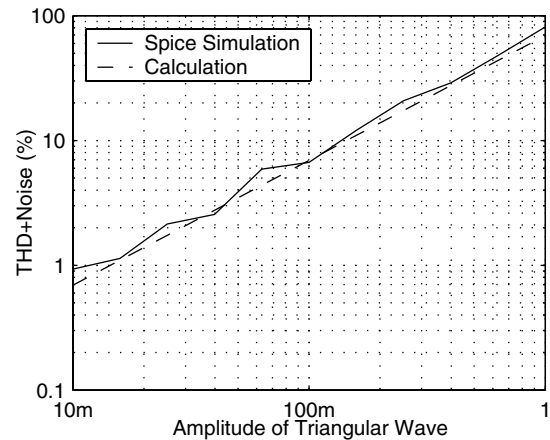


Fig. 6. THD+Noise vs. Triangular Wave Amplitude [$\Delta t = 1ns$, $V_{SP} = 0.5V$, $T_{tri} = 1\mu s$, $f_{in} = 1KHz$, $f_{bw} = 15 KHz$]

is shown in Fig. 6. In this simulation the input signal amplitude is set to 1 mV and triangular wave amplitude is swept from 10 mV to 1 V to avoid clipping. The simulation results show good agreement with the theoretical derivations.

THD+Noise is independent of the input signal frequency as expected. The result of THD+Noise vs. input frequency is shown in Fig. 7. However, due to the nonlinear inductance, THD is a function of input frequency. To test the effect of the nonlinear inductor, a second order LC filter with 20KHz cut-off is connected to the output of the amplifier. Input frequency vs. THD is shown in Figure 8. At low frequencies, the simulation results show very good agreement with the calculation results. Near the cut-off frequency, THD starts to decrease, because of the assumption of the derivation, therefore, the error between the model and real circuit is increased. Although the model can be improved using Volterra series analysis, this will result in more complex equation, which is not practical.

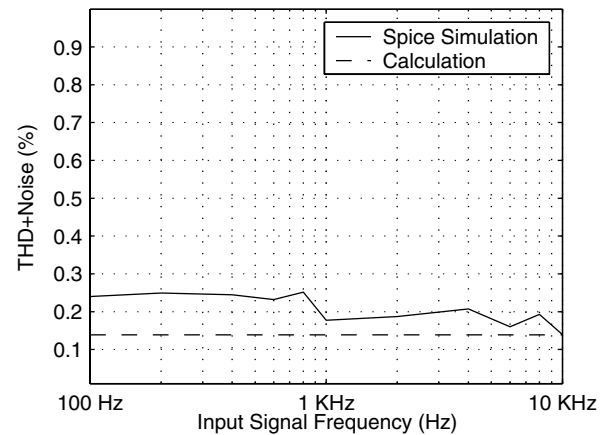


Fig. 7. THD+Noise vs. Input Signal Frequency [$\Delta t = 1ns$, $V_{SP} = 0.5V$, $T_{tri} = 1\mu s$, $V_{tri} = 1V$, $f_{bw} = 15 KHz$]

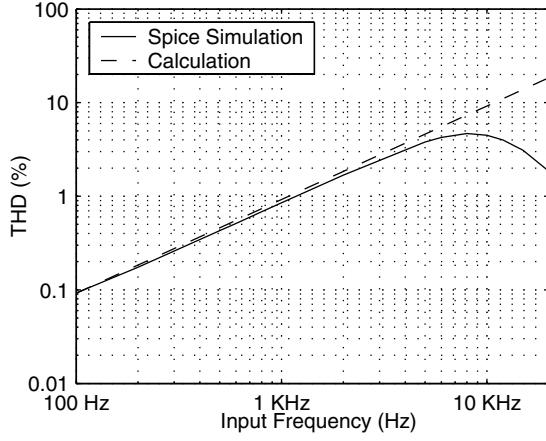


Fig. 8. THD vs. Input Signal Frequency [$\Delta t = 1ns$, $V_{tri} = 1V$, $V_{SP} = 0.5V$, $T_{tri} = 1\mu s$, $V_{tri} = 1V$, $f_{bw} = 15 KHz$]

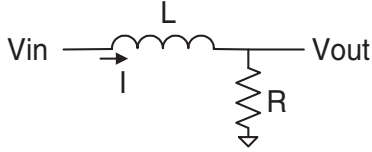


Fig. 9. Equivalent Low-Pass Filter below Cutoff Frequency

IV. CONCLUSION

The effect of two important parameters of class-D amplifier on THD+Noise is investigated: the jitter at the edges of PWM signal and nonlinear inductor. A mathematical analysis is presented to model the relationship among these parameters and THD+Noise. The proposed mathematical analysis provides an insight to the THD+Noise of a class-D amplifier. The accuracy of the analysis is also verified by transient simulations at transistor level. The simulation results support the accuracy of the mathematical model.

V. APPENDIX

At lower frequencies than cutoff frequency of the low-pass filter, the capacitance in Fig. 1 is negligible, therefore the filter is simplified as in Fig. 9, where R is the resistance of the speaker. Since total harmonic distortion is measured for sinusoidal input, the input signal is

$$V_{in} = V_o \sin(\omega_{in} t) \quad (8)$$

For small input signal levels and weakly nonlinearity, input and output voltages is assumed same for frequencies lower than cut-off frequency of the filter. Therefore, the inductor current is

$$I = \frac{V_{out}}{R} \approx \frac{V_{in}}{R} \quad (9)$$

The error due to this assumption is

$$V_{err} = V_{in} - V_{out} = \frac{d}{dt}[L \cdot I] \quad (10)$$

Inserting Eqs. 6, 8 and 9 into Eq. 10, and arranging the equation, the error voltage is

$$V_{err} = \left[\frac{L_0}{R} V_o \omega_{in} + \frac{L_0 a_3}{R^3} V_o^3 \frac{3\omega_{in}}{4} \right] \cos(\omega_{in} t) - \frac{L_0 a_3}{R^3} V_o^3 \frac{3\omega_{in}}{4} \cos(3\omega_{in} t) \quad (11)$$

The output voltage is only the difference between the input voltage and error voltage. The first part of Eq. 11 is also at the signal frequency and much smaller than the input signal level for the band of interest. Therefore, it is negligible. The output voltage is

$$V_{out} = V_{in} - V_{err} = V_o \sin(\omega_{in} t) - \frac{L_0 a_3}{R^3} V_o^3 \frac{3\omega_{in}}{4} \cos(3\omega_{in} t) \quad (12)$$

The third-order harmonic distortion is easily obtained by using Eq. 12.

$$HD3 = \frac{\frac{L_0 a_3}{R^3} V_o^3 \frac{3\omega_{in}}{4}}{V_o} = \frac{3 \cdot L_0 V_o^2 0.1 \cdot 2\pi f_{in}}{4 \cdot R^3 I_{sat}^2} \quad (13)$$

Since third harmonic is the only harmonic the total harmonic distortion simply equals HD3.

REFERENCES

- [1] J. W. Lee, J. S. Lee, G. S. Lee, and S. Kim, "A 2W BTL Single-Chip Class-D Power Amplifier with Very High Efficiency for Audio Applications," *Proc. IEEE ISCAS*, vol. V, pp. 493-496, 2000.
- [2] P. Dondon, and J. M. Micoulet, "An Original Approach for The Design of A Class D Power Switching Amplifier-An Audio Application," *Proc. IEEE ICECS*, vol. 1, pp. 161-164, 1999.
- [3] R. E. Hiorns, J. M. Goldberg, M. B. Sandler, "Design Limitations for Digital Audio Power Amplification," *Digital Audio Signal Processing, IEEE Colloquium on*, pp. 4/1-4/4, May 22, 1991.
- [4] F. A. Himmelstoss, K. H. Edelmöser, and C. C. Anselmi, "Analysis of A Quality Class-D Amplifier," *IEEE Trans. on Consumer Electronics*, Vol. 42, No. 3, pp. 860-864, Aug. 1996.
- [5] M. T. Tan, H. C. Chua, B. H. Gwee, J. S. Chang, "An Investigation on the Parameters Affecting Total Harmonic Distortion in Class D Amplifiers," *Proc. IEEE ISCAS*, Vol. 4, pp. 193-196, May 28-31, 2000.
- [6] D. Middleton, *An Introduction to Statistical Communication Theory*, McGraw Hill, New York, pp. 218-227, 1960.
- [7] A. M. Stanković, G. C. Verghese, D. J. Perreault, "Analysis and Synthesis of Randomized Modulation Schemes for Power Converters," *IEEE Tran. on Power Electronics*, Vol. 10, No. 6, Nov. 1995.
- [8] K. Kundert, "Modeling Jitter in PLL-based Frequency Synthesizers," [Online] Available: <http://www.designers-guide.com/Analysis/PLLnoise+jitter.pdf>.
- [9] M. Shimanouchi, "An approach to consistent jitter modeling for various jitter aspects and measurement methods," *Proc. ITC International Test Conference 2001*, Pages 848 - 857, 30 Oct. - 1 Nov. 2001.
- [10] H. S. Black, *Modulation Theory*, Van Nostrand, Princeton, NJ, pp. 263-281, 1953.
- [11] H. S. Rowe, *Signals and Noise in Communication Systems*, Van Nostrand, Princeton, NJ, pp. 257-310, 1965.
- [12] M. Bazes, "Two Novel Fully Complementary Self-Biased CMOS Differential Amplifiers," *IEEE J. Solid-State Circuits*, Vol. 26, No. 2, Feb. 1991.